Understand, try, revisit

1.1 Index, value and update

First use: S2 and S4.

Definition · A sequence is not a finite list

The index n denotes an integer; u_n is a real number. The representative point is $(n; u_n)$. A line connecting the points can help reading but does not define terms with non-integer indices.

Worked example · Two expressions that look alike

For $u_0 = 2$ and $u_{n+1} = 3u_n - 1$, we get $u_1 = 5$, $u_2 = 14$, $u_3 = 41$. Thus $u_2 + 1 = 15$ while that $u_{2+1} = 41$. In a loop, the variable `u` contains first u_n; the instruction `u = 3*u-1` replaces it with u_{n+1}.

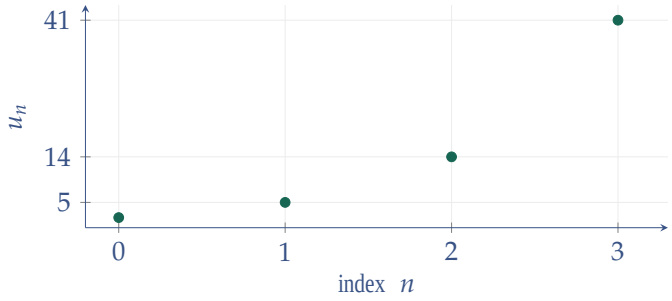


Figure 1.1. Read the term of index 2: its y-coordinate is 14, not 2.

A1 · Immediate application For $v_0 = 1$, $v_{n+1} = 2v_n + 1$, build a table of indices 0 to 3, then place the four points. Calculate separately $v_2 + 1$ and v_{2+1}.

A2 · Partial guidance A loop must calculate $u_0 + u_1 + u_2 + u_3$ for the rest of the example. It starts of `u=2; total=2`. How many updates are needed? Write the order of the instructions and explain why adding before the update would count twice u_0.

1.2 A local hypothesis, a universal conclusion

First use: S5 and S7.

Worked example · Induction for an inequality

Let $w_0 = 1$ and $w_{n+1} = \frac{1}{2}w_n + 2$. Let us prove $P(n) : 1 \leq w_n < 4$. Initially $w_0 = 1 \in [1; 4[$. Fix $n \in \mathbb{N}$ and suppose $P(n)$. Since $\frac{1}{2} > 0$, $\frac{1}{2} \leq \frac{1}{2}w_n < 2$, so $\frac{5}{2} \leq w_{n+1} < 4$. Since $\frac{5}{2} \geq 1$, this implies $P(n+1)$. The principle of induction gives the bounds at every index.

A3 · Proof to be completed For $a_0 = 2$, $a_{n+1} = 2a_n + 1$, complete: " $P(n)$ means $a_n = 3 \times 2^n - 1$. At index 0, . . . Fix . . . and assume . . . Then $a_{n+1} = \ldots = 3 \times 2^{n+1} - 1$. Hence . . ."

A4 · Fix a proof A student writes: "If $u_n \geq 2$, then $u_{n+1} = 2u_n \geq 4 \geq 2$. So all the terms are at least 2." He did not give u_0. Construct a counterexample, then state which given makes its proof complete.

1.3 Count before summing

First use: S10.

Property · Number of terms

For integers $p \leq n$, the sum $\sum_{k=p}^{n} u_k$ contains $n - p + 1$ terms. This can be checked by writing out the first and last ones, before any formula.

Worked example · A shifted geometric sum

In $\sum_{k=3}^{7} 2^k$, there are five terms and the first equals 8. The sum is $8 \frac{1-2^5}{1-2} = 248$. For prove the formula without memorising it in isolation, write $S = 1 + q + \cdots + q^n$, then subtract $qS = q + \cdots + q^{n+1}$: $(1-q)S = 1 - q^{n+1}$. Dividing assumes $q \neq 1$.

A5 · Independent proof Prove by induction, for every $n \in \mathbb{N}$, $\sum_{k=0}^{n}(2k+1) = (n+1)^2$. Explain why the term added in the inductive step is 2 $n + 3$, and not $2n + 1$.

1.4 Recognise a structure without confusing it with a limit

First use: S11 and S13.

Worked example · Choosing the shift

For $w_{n+1} = \frac{1}{2}w_n + 2$, the fixed value is 4. Subtracting 4 from both sides, $w_{n+1} - 4 = \frac{1}{2}(w_n - 4)$. Thus $w_n = 4 - 3\,2^{-n}$. The shift is not arbitrary: it removes the term constant. This explicit calculation does not yet assume a convergence theorem.

A6 · Choice of strategy A subscription costs 40 euros in the first year, then its cost increases each year by 5 euros. Another costs 40 euros and then increases by 10 % per year. Write both sequences, compare the first three years and calculate the cumulative costs over ten years. Justify the number of terms.

A first step: First name the type of the two changes. The tenth year corresponds to index 9 if the first is index 0.

One minute, spoken aloud. Explain the difference between "for every n, $u_n \leq 4$" and "there exists n such that $u_n \leq 4$". Give a sequence satisfying the second but not the first.

Delayed review · S20 then S41. Revisit $w_n = 4 - 3\,2^{-n}$ to prove its convergence; in probabilities, look again for the shift when a probability satisfies an affine relation.

Investigation

A tank topped up every day - 35 min.

A tank initially contains 100 litres of water. Each day, 20 % of the volume present is removed, then 30 litres are added. For every natural n, let u_n the volume of water, in litres, after these two operations performed n times. Thus, $u_0 = 100$.

1. Calculate u_1 and u_2, then express u_{n+1} as a function of u_n.
2. Which value of the volume would remain unchanged after one day? Prove by induction that, for all $n \in \mathbb{N}$, $100 \leq u_n < 150$.

3. For every $n \in \mathbb{N}$, set $v_n = u_n - 150$. Show that (v_n) is geometric. Deduce the expressions for v_n and u_n as a function of n.

4. Write a Python function with no argument that returns the first integer n for which $u_n > 145$. Run the program and interpret the result.

5. A student calculated the first hundred terms and observes that they are less than 150. This observation enough to prove that all the terms are? Justify.